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-- 1.
select e.emp_id, e.emp_name, e.dept_id, e.salary from employees e where e.salary >=
(select avg(emp.salary) from employees emp where emp.dept_id = e.dept_id);

-- 2.
select emp_id, emp_name, dept_id, salary
from employees e
where salary = (select max(salary) from employees where dept_id = e.dept_id);

-- 3.
select max(salary) from employees where salary < (select max(salary) from employees);

-- 4.
select emp_id, salary from employees where salary in (select max(salary) from employees
where salary < (select max(salary) from employees));

-- 5.
select emp_id, emp_name, salary, manager_id
from employees e
where salary > (select salary from employees where emp_id = e.manager_id);

-- 6.
select emp_id, emp_name, salary
from employees e
where exists (select 1 from employees r where r.manager_id = e.emp_id and r.salary >
e.salary);

-- 7.
select dept_id
from employees
group by dept_id
having avg(salary) > (select avg(salary) from employees);

-- 8.
select emp_id, emp_name, dept_id, salary
from employees
where dept_id in (select dept_id from employees group by dept_id having avg(salary) >
(select avg(salary) from employees));

-- 9.
select emp_id, emp_name, salary
from employees
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where salary > all (select salary from employees where dept_id = (select dept_id from departments where dept_name = 'HR'));
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-- 10.

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select emp_id, emp_name, salary
from employees
where salary > any (select salary from employees where dept_id = (select dept_id from departments where dept_name = 'Research'));
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-- 11.

```
select dept_id, dept_name
from departments d
where not exists (select 1 from projects p where p.dept_id = d.dept_id);
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-- 12.

```
select emp_id, emp_name, dept_id
from employees
where dept_id in (select dept_id from projects group by dept_id having count(*) > 1);
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-- 13.

```
select emp_id, emp_name, dept_id, salary
from employees
where dept_id in (select dept_id from projects where budget = (select max(budget) from projects));
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-- 14.

```
select emp_id, emp_name, salary
from employees
where abs(salary - (select avg(salary) from employees)) = (select min(abs(salary - (select avg(salary) from employees))) from employees);
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-- 15.

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select dept_id
from employees e
group by dept_id
having max(salary) > all (select salary from employees m where m.dept_id = e.dept_id and m.emp_id in (select manager_id from employees where manager_id is not null));
```